

Nathan Zhao

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Education

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Electrical Engineering (Circuit Technology, Robotics)

Expected December 2029

The Pembroke Hill School

Kansas City, Missouri

High School Diploma, Cum Laude

• **Honors & Coursework:** ACT 36 · GPA 97.7 · Multivariable Calculus, Differential Equations

Work Experience

Founding Engineer · Effluent

2026

- Designed and fabricated gas sensor array carrier boards and wrote C++ microcontroller firmware integrating open62541 and MQTT to bridge edge sensor data telemetry to automation and analytics systems.
- Built simulation tools, Python data pipelines, and a cross-platform C#/.NET GUI delivering actionable insights interpreted from WWTP scenario modeling directly to plant operators.
- Designed ML architecture fusing sparse attention with dilated inception graph convolutional networks for time-series forecasting of wastewater process data, with emphasis on per-facility transfer learning.

Research & Development Intern · MRIGlobal

2025 – 2026

- Led development of portable, low-cost chemically-functionalized gas sensor arrays, including sensor fabrication, controlled testing, and signal processing, for real-time detection and classification of chemical warfare agents.
- Built Swagelok vacuum-oven vapor-deposition system and sensor-characterization setups with closed-loop analyte gas mass-flow control software and a custom human-machine interface.
- Designed and fabricated a device PCB (Altium) and deployed a CNN-BiLSTM model on-device (TensorFlow Lite), reaching >96% preliminary classification accuracy on edge inference.

Research Assistant · UMKC Multiplatform Interactive Robotics Lab (mirolab)

2025 – 2026

- Led development of vision-based hand-joint goniometry system using multi-view MediaPipe hand pose estimation with FDM 3D-printed ArUco marker glove via nonlinear Kalman fusion, reaching sub-degree error in controlled flexion-angle measurement validation.
- Co-authoring manuscript intended for submission to *IEEE Robotics and Automation Letters* (PI: Hee Sup Shin).

Research Intern · University of Pittsburgh, Accessible SmartTech Lab

2024

- Developed autonomous object-handoff behaviors on a Hello Robot Stretch RE1 (ROS Noetic) using ArUco markers, YOLOv5 object detection, and 2D LiDAR SLAM (PI: Dan Ding).

Selected Projects

Bimanual Mobile Manipulator for Assisted Living

2024 – Present

- Dual telescopic-arm mobile robot for in-home elderly-care tasks. *Onshape*, C++, *PlatformIO*.

mmHome: Multimodal IoT Contactless Vital-Sign Monitoring & Fall Detection

2025 – 2026

- Wall-mounted mmWave + vibration sensing for respiration and fall detection. *Python*, *scikit-learn*, C++.

LeSCARA: Open-Source SCARA Platform for Embodied AI Research

2026 – Present

- Low-cost 3D-printed SCARA arm for reproducible learning-based manipulation. *Onshape*, C++, *LeRobot*.

Relevant Skills

Computer Aided Design: Onshape, SolidWorks, KiCAD, Altium Designer

Programming: C++, Python, C#/.NET, Java, MATLAB, Linux, Docker, Git

ML & Vision: PyTorch, OpenCV, scikit-learn, TensorFlow Lite, pandas

Robotics & Embedded: ROS/ROS2, MuJoCo, PyBullet, PlatformIO, PicoSDK, Arduino

Languages: English, Chinese (Mandarin), Spanish